Tutorial 3 - Solutions

1. (a) To find the MME, note that $\mu(p) = 1/p$. Therefore, set $\bar{X} = 1/\tilde{p}$ and solve for $\tilde{p}$ to obtain MME: $\tilde{p} = 1/\bar{X}$.

To find the MLE, maximize the log-likelihood function:

$$\text{Likelihood function} \quad : \quad L(p;\underline{x}) = \prod_{i=1}^{n} p(1-p)^{x_i-1} = p^n(1-p)^{\Sigma x_i - n}$$

$$\text{Log-likelihood function} \quad : \quad \log L(p;\underline{x}) = n\log p + (\sum_{i=1}^{n} x_i - n)\log(1-p)$$

$$\text{Differentiating} \quad : \quad \frac{d}{dp}\log L(p;\underline{x}) = \frac{n}{p} - \frac{\sum_{i=1}^{n} x_i - n}{1-p}.$$

Solve $\frac{d}{dp}\log L(p;\underline{x}) = 0$ to get MLE: $\hat{p} = 1/\bar{X}$

Log-likelihood is *maximized* at $\hat{p}$, since $\frac{d^2}{d^2p}\log L(p;\underline{x}) = \frac{-n}{p^2} - \frac{\Sigma x_i - n}{(1-p)^2} < 0$, since $\Sigma x_i \geq n$ always.

(b) To find the MME, note that $\mu(\lambda) = t/\lambda$ (recall that $t > 0$ is *known*). Therefore, set $\bar{X} = t/\tilde{\lambda}$ and solve for $\tilde{\lambda}$ to obtain MME: $\tilde{\lambda} = t/\bar{X}$.

To find the MLE, maximize the log-likelihood function:

$$\text{Likelihood} \quad : \quad L(\lambda;\underline{x}) = \prod_{i=1}^{n} \frac{x_i^{t-1}\lambda^t e^{-\lambda x_i}}{\Gamma(t)} = \frac{(x_1 \cdots x_n)^{t-1}\lambda^{nt}e^{-\lambda\Sigma x_i}}{(\Gamma(t))^n}$$

$$\text{Log-likelihood} \quad : \quad \log L(\lambda;\underline{x}) = (t-1)\sum_{i=1}^{n}\log x_i + nt\log(\lambda) - \lambda\sum_{i=1}^{n} x_i - n\log\Gamma(t)$$

$$\text{Differentiating} \quad : \quad \frac{d}{d\lambda}\log L(\lambda;\underline{x}) = \frac{nt}{\lambda} - \sum_{i=1}^{n} x_i.$$

Solve $\frac{d}{d\lambda}\log L(\lambda;\underline{x}) = 0$ to get MLE: $\hat{\lambda} = t/\bar{X}$

Log-likelihood is *maximized* at $\hat{\lambda}$, since $\frac{d^2}{d^2\lambda}\log L(\lambda;\underline{x}) = \frac{-nt}{\lambda^2} < 0$.

(c) To find the MME, note that $\mu(\theta) = \frac{\theta}{\theta+1}$ (check this by computing the expectation directly). Therefore, set $\bar{X} = \frac{\tilde{\theta}}{\tilde{\theta}+1}$ and solve for $\tilde{\theta}$ to get MME: $\tilde{\theta} = \bar{X}/(1-\bar{X})$.

To find the MLE, maximize the log-likelihood function:

$$\text{Likelihood} \quad : \quad L(\theta;\underline{x}) = \prod_{i=1}^{n} \theta x_i^{\theta-1} = \theta^n(x_1x_2\cdots x_n)^{\theta-1}$$

$$\text{Log-likelihood} \quad : \quad \log L(\theta;\underline{x}) = n\log\theta + (\theta-1)\sum_{i=1}^{n}\log(x_i)$$

$$\text{Differentiating} \quad : \quad \frac{d}{d\theta}\log L(\theta;\underline{x}) = \frac{n}{\theta} + \sum_{i=1}^{n}\log(x_i).$$

Solve $\frac{d}{d\theta}\log L(\theta;\underline{x}) = 0$ to get MLE: $\hat{\theta} = -n/\Sigma\log(X_i)$.

Log-likelihood is *maximized* at $\hat{\theta}$, since $\frac{d^2}{d^2\theta}\log L(\theta;\underline{x}) = -n/\theta^2 < 0$.

2. (a) Since observations are independent we have that

$$L(\lambda;(3,6,2)) = \frac{e^{-\lambda}\lambda^3}{3!} \times \frac{e^{-\lambda}\lambda^6}{6!} \times \frac{e^{-\lambda}\lambda^2}{2!} = \frac{e^{-3\lambda}\lambda^{11}}{8640}$$

The log of the likelihood is given by

$$l(\lambda;(3,6,2)) = 11\log\lambda - 3\lambda - \log 8640.$$

(b) Differentiate, equate to zero, solve and check for maximum to obtain the MLE of λ as $\hat{\lambda} = \frac{11}{3} = 3.67$.

(c) For the Poisson(λ) model the population mean is equal to λ. We obtain the method of moments estimate by equating the sample mean to population mean and solving for λ. This gives the same estimate as (b).

3. (a) Note that we do not have observations from an exponential distribution. We observe whether a light bulb has failed by a certain time t_0. Thus we have n observations from Bin$(1,p)$ where $p = \mathbb{P}(\text{bulb fails by time } t_0) = 1 - e^{-\lambda t_0}$. In other words, we have an observation from Bin$(n, 1-e^{-\lambda t_0})$.

(b) For the MME, the mean of X is np so we set

$$x = np = n(1-e^{-\lambda t_0}) \;\Rightarrow\; \hat{\lambda} = -t_0^{-1}\log(1-x/n).$$

(c) The likelihood function is

$$L(\lambda) \propto (1-e^{-\lambda t_0})^x (e^{-\lambda t_0})^{n-x}.$$

Taking logs gives

$$\ell(\lambda) = \log L = x\log(1-e^{-\lambda t_0}) - (n-x)\lambda t_0.$$

Differentiating etc., we find that $\hat{\lambda} = -t_0^{-1}\log(1-x/n)$, the same as the MME.

4. (a) Let $F(x) = \mathbb{P}\{X_{(1)} \le x\} = 1 - \mathbb{P}\{X_{(1)} > x\}$ be the distribution function for $X_{(1)} = \min\{X_1, X_2, ..., X_n\}$, where $X_1, X_2, ..., X_n$ are iid Exp$(1/\theta)$ random variables. Then

$$\begin{aligned}
\mathbb{P}\{X_{(1)} > x\} &= \mathbb{P}\{X_1 > x, X_2 > x, ..., X_n > x\} \\
&= \mathbb{P}\{X_1 > x\}\mathbb{P}\{X_2 > x\}\cdots\mathbb{P}\{X_n > x\} \\
&\qquad \text{since } X_1, ..., X_n \text{ are independent} \\
&= (\mathbb{P}\{X_1 > x\})^n \quad \text{since } X_1, ..., X_n \text{ are identically distributed} \\
&= (e^{-x/\theta})^n = e^{-nx/\theta} \quad \text{since } X_1 \sim \text{Exp}(1/\theta).
\end{aligned}$$

It follows that $X_{(1)} \sim \text{Exp}(n/\theta)$ with mean $\frac{\theta}{n}$. Hence $\mathbb{E}(g(\underline{X}) = \mathbb{E}(nX_{(1)}) = n \times \frac{\theta}{n} = \theta$, i.e. $nX_{(1)}$ is an unbiased estimator for θ.

(b) Since $nX_{(1)}$ is an unbiased estimator for θ, we have

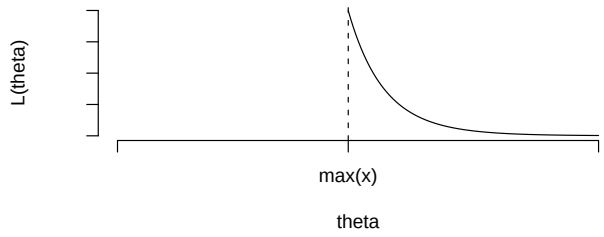
$$\text{MSE}(nX_{(1)}) = \mathbb{V}\text{ar}(nX_{(1)}) = n^2\mathbb{V}\text{ar}(X_{(1)}) = n^2 \times \left(\frac{\theta}{n}\right)^2 = \theta^2$$

since $X_{(1)} \sim \text{Exp}(n/\theta)$.

5. (a) The likelihood function $L(\theta;\underline{x})$ is given by

$$L(\theta;\underline{x}) = \begin{cases} \prod_{i=1}^{n} f(x_i;\theta) = \frac{2^n x_1 x_2 \cdots x_n}{\theta^{2n}} & \text{if} \quad \theta \geq x_{(n)} = \max\{x_1, x_2, ..., x_n\} \\ 0 & \text{if } \theta < x_{(n)}. \end{cases}$$

Sketch of $L(\theta;\underline{x})$:



We see from the graph of $L(\theta;\underline{x})$ that the MLE for θ is given by $X_{(n)}$.

(b) For $0 < x < \theta$, we have

$$\begin{aligned} F_{X_{(n)}}(x) &= \mathbb{P}\{X_{(n)} \leq x\} \\ &= \mathbb{P}\{X_1 \leq x, X_2 \leq x, ..., X_n \leq x\} \\ &= \mathbb{P}\{X_1 \leq x\}\mathbb{P}\{X_2 \leq x\} \cdots \mathbb{P}\{X_n \leq x\} \\ &\qquad \text{since } X_1, ..., X_n \text{ are independent} \\ &= (\mathbb{P}\{X_1 \leq x\})^n \quad \text{since } X_1, ..., X_n \text{ are identically distributed} \\ &= \left(\frac{x^2}{\theta^2}\right)^n = \frac{x^{2n}}{\theta^{2n}}. \end{aligned}$$

Also, $F_{X_{(n)}}(x) = 0$ when $x \leq 0$, and $F_{X_{(n)}}(x) = 1$ when $x \geq \theta$. It follows that the density function for $X_{(n)}$ is given by:

$$f_{X_{(n)}}(x) = \begin{cases} \frac{2nx^{2n-1}}{\theta^{2n}} & \text{if} \quad 0 < x < \theta, \\ 0 & \text{otherwise.} \end{cases}$$

Hence

$$\mathbb{E}(X_{(n)}) = \int_0^\theta \frac{2nx^{2n}}{\theta^{2n}} dx = \frac{2n\theta}{2n+1}.$$

Also,

$$\mathbb{E}(X_{(n)}^2) = \int_0^\theta \frac{2nx^{2n+1}}{\theta^{2n}} dx = \frac{n\theta^2}{(n+1)},$$

and so

$$\mathbb{V}\text{ar}(X_{(n)}) = \mathbb{E}(X_{(n)}^2) - (E(X_{(n)}))^2 = \frac{n\theta^2}{(n+1)} - \left(\frac{2n\theta}{2n+1}\right)^2$$

$$= \frac{n\theta^2}{(n+1)(2n+1)^2}.$$

(c) Since $\mathbb{E}(X_{(n)}) = \frac{2n\theta}{2n+1}$, for $\gamma = \frac{2n+1}{2n}$ the estimator $\gamma X_{(n)}$ is *unbiased*.
Check:

$$\mathbb{E}\left(\frac{2n+1}{2n}X_{(n)}\right) = \frac{2n+1}{2n}\mathbb{E}(X_{(n)}) = \theta.$$

So:

$$\begin{aligned}
\text{MSE}\left(\frac{2n+1}{2n}X_{(n)}\right) &= \mathbb{V}\text{ar}\left(\frac{2n+1}{2n}X_{(n)}\right) \\
&= \left(\frac{2n+1}{2n}\right)^2 \mathbb{V}\text{ar}(X_{(n)}) \\
&= \left(\frac{2n+1}{2n}\right)^2 \frac{n\theta^2}{(2n+1)^2(n+1)} \\
&= \frac{\theta^2}{4n(n+1)}.
\end{aligned}$$

In Tutorial 2, Question 8 you have shown that $\text{MSE}(\frac{3\bar{X}}{2}) = \frac{\theta^2}{8n}$. Now, observe that for $n \geq 2$

$$\text{MSE}\left(\frac{2n+1}{2n}X_{(n)}\right) = \frac{\theta^2}{4n(n+1)} < \frac{\theta^2}{8n} = \text{MSE}(\hat{\theta}).$$

So $\frac{2n+1}{2n}X_{(n)}$ is a (much) more efficient unbiased estimator than $\hat{\theta} = \frac{3\bar{X}}{2}$.

6. If $X \sim \text{Poi}(\lambda)$ it is straightforward to show (e.g. see lecture notes) that the MLE for λ is given by the sample mean, i.e. $\hat{\lambda} = \bar{X}$. Here we have $\bar{x} = \frac{8}{10} = 0.8 \Rightarrow \hat{\lambda} = 0.8$.

We want the MLE for the probability $\varphi = \mathbb{P}(X \geq 2)$. First notice that

$$\begin{aligned}
\varphi &= 1 - \mathbb{P}(X < 2) = 1 - \mathbb{P}(X = 0 \cup X = 1) = 1 - \{\mathbb{P}(X = 0) + \mathbb{P}(X = 1)\} \\
&= 1 - (e^{-\lambda} + e^{-\lambda}\lambda) = 1 - e^{-\lambda}(1 + \lambda).
\end{aligned}$$

Then the invariance property of the MLE gives the MLE of φ as

$$\hat{\varphi} = 1 - e^{-\hat{\lambda}}(1 + \hat{\lambda}) = 0.191.$$

7. (a) Let X be the time till relapse (in months) for a given patient then $X \sim \text{Gamma}\,(2, \beta)$. The probability density function for a Gamma $(2, \beta)$ random variable is

$$f_X(x) = \frac{\beta^2}{\Gamma(2)}xe^{-\beta x}, \quad \text{for } x \geq 0.$$

Recall that $\Gamma(n) = (n-1)!$ for any positive integer n so $\Gamma(2) = 1$. Hence the probability that a patient doesn't relapse in the 6-month trial is given by $\mathbb{P}(X > 6)$ which we can calculate using the probability density function as

$$\mathbb{P}(X > 6) = 1 - \int_{-\infty}^{6} f_X(x)dx = 1 - \int_0^6 \beta^2 xe^{-\beta x}dx.$$

Integration by parts gives:

$$\begin{aligned}
\int_0^6 \beta^2 xe^{-\beta x}dx &= \beta\left[-xe^{-\beta x}\right]_{x=0}^{x=6} + \beta\int_0^6 e^{-\beta x}dx \\
&= \beta\left[-xe^{-\beta x}\right]_{x=0}^{x=6} + \left[-e^{-\beta x}\right]_{x=0}^{x=6} \\
&= -6\beta e^{-6\beta} + 1 - e^{-6\beta} = 1 - (1 + 6\beta)e^{-6\beta}
\end{aligned}$$

Therefore

$$1 - \int_0^6 \beta^2 x e^{-\beta x} dx = (1 + 6\beta)e^{-6\beta}.$$

(b) In this case we have censored data so we need to distinguish between the data we observe $x_i < 6$ and the remaining ones that are not observed. The likelihood is given by

$$L(\beta; \underline{x}) = \prod_{i=1}^{10} f_X(x_i; \beta).$$

For the 6 patients who relapse within 6 months they contribute the following to the likelihood:

$$\prod_{i=1}^{6} f_X(x_i; \beta) = \prod_{i=1}^{6} \beta^2 x_i e^{-\beta x_i} = \beta^{12} e^{-\beta \sum_{i=1}^{6} x_i} \prod_{i=1}^{6} x_i$$

We are given $\sum_{i=1}^{6} = 13.2$, so

$$\prod_{i=1}^{6} f_X(x_i; \beta) = \beta^{12} e^{-13.2\beta} \prod_{i=1}^{6} x_i$$

For the patients that we didn't observe we use the probability that they didn't relapse within 6 months for the likelihood, which we calculated in part (a). Indeed, these 4 patients contribute the following to the likelihood:

$$\prod_{i=7}^{10} \mathbb{P}(X_i > 6; \beta) = \left((1 + 6\beta)e^{-6\beta}\right)^4 = (1 + 6\beta)^4 e^{-24\beta}$$

Combining the contributions of all patients gives:

$$L(\beta; \underline{x}) = \left(\beta^{12} e^{-13.2\beta} \prod_{i=1}^{6} x_i\right) \left((1 + 6\beta)^4 e^{-24\beta}\right) = \beta^{12}(1 + 6\beta)^4 e^{-37.2\beta} \prod_{i=1}^{6} x_i$$

For the purposes of maximising this function we can treat $\prod_{i=1}^{6} x_i$ as a constant (in β) and write

$$L(\beta; \underline{x}) \propto \beta^{12}(1 + 6\beta)^4 e^{-37.2\beta}$$

(c) To find the MLE of β, we take the log of the likelihood:

$$l(\beta; \underline{x}) = 12 \log \beta + 4 \log(1 + 6\beta) - 37.2\beta$$

Now differentiate and equate to zero to give

$$\frac{12}{\beta} + \frac{24}{1 + 6\beta} - 37.2 = 0.$$

On multiplying through by $-\beta(1 + 6\beta)$ we obtain

$$223.2\beta^2 - 58.8\beta - 12 = 0.$$

This is solved by $\beta = -0.13, 0.40$. Since β must be positive the MLE is $\hat{\beta} = 0.40$.

Exam Style Question

(a) First we calculate the expected value of X as a function of θ. By definition

$$\mathbb{E}[X] = \int_\theta^\infty x e^{\theta - x} dx.$$

Using the substitution $y = x - \theta$ we have

$$\mathbb{E}[X] = \int_0^\infty (y + \theta) e^{-y} dy.$$

Note that e^{-y} is the probability density function of Exp(1) so

$$\int_0^\infty e^{-y} dy = 1$$
$$\int_0^\infty y e^{-y} dy = 1$$

and hence

$$\mathbb{E}[X] = \int_0^\infty y e^{-y} dy + \theta \int_0^\infty e^{-y} dy = 1 + \theta.$$

Rearranging we have

$$\theta = \mathbb{E}[X] - 1$$

and estimating $\mathbb{E}[X]$ with $\bar{X}$ gives the MME:

$$\hat{\theta}(\underline{X}) = \bar{X} - 1$$

[5 marks]

(b) The likelihood is

$$L(\theta; \underline{x}) = \prod_{i=1}^n f(x_i; \theta) = \prod_{i=1}^n e^{\theta - x_i} = \exp\left(\sum_{i=1}^n (\theta - x_i) \right)$$

provided $\theta \leq x_i$ for all i. Otherwise $L(\theta; \underline{x}) = 0$. The log-likelihood is

$$\ell(\theta; \underline{x}) = \log(L(\theta; \underline{x})) = \sum_{i=1}^n (\theta - x_i) = n(\theta - \bar{x})$$

provided $\theta \leq x_i$ for all i. Otherwise $\ell(\theta; \underline{x}) = -\infty$.

[2 marks]

(c) The maximum likelihood estimator is defined as

$$\hat{\theta} = \arg\min_\theta \ell(\theta; \underline{x}).$$

Note that $\ell(\theta; \underline{x}) = -\infty$ unless $\theta \leq \min(x_1, \ldots, x_n)$ so using our expression for $\ell(\theta; \underline{x})$

$$\hat{\theta} = \arg\min_{\theta \leq \min(\underline{x})} n(\theta - \bar{x}).$$

Since $n(\theta - \bar{x})$ is increasing with θ, ℓ takes its largest value at the largest value of θ within the constraints. That is, $\ell(\theta; \underline{x})$ is maximised when $\theta = \min(x_1, \ldots, x_n)$.

[2 marks]

[Total 9 marks]